

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO-2 : Apply CSS layout and styling techniques to create visually appealing and responsive web interfaces, and use JavaScript to enable interactive client-side behavior. (L3)

Assessment Tool : Short Answer Test

Weightage: 10%

QUESTIONS:

Total Marks: 20

Question 1 (5 Marks)

Suppose that a company wants its website to work seamlessly on mobile, tablet, and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox or Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

Question 2 (5 Marks)

Suppose that a webpage contains a form where user input must be validated before submission. Explain the concept of client-side validation and its purpose. Describe how JavaScript event handling is used in validation and explain the role of regular expressions in validating inputs such as email or phone numbers. Provide a suitable example and suggest improvements for better usability and security.

Question 3 (5 Marks)

Suppose that a webpage layout breaks when the browser window is resized. Explain the CSS box model and its components, and describe how CSS positioning techniques affect layout structure. Identify possible causes of layout breaking and explain how media queries can be used to resolve these issues. Suggest best practices to ensure a stable and responsive layout.

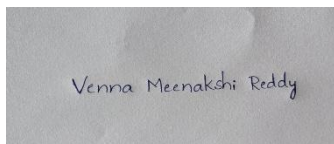
Question 4 (5 Marks)

Suppose that a website requires dynamic content updates without reloading the entire page. Explain the concept of the Document Object Model (DOM) and its role in web development. Describe how JavaScript can be used to manipulate the DOM for updating webpage content dynamically. Provide an example and suggest improvements to enhance performance and user experience.

Code of Conduct:

I VENNA MEENAKSHI REDDY (192471048) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

A photograph of a piece of paper with the handwritten signature "Venna Meenakshi Reddy" in blue ink.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
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Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with	Incorrect or irrelevant answer
				noticeable errors	
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

CO2 - AT3

- 1) Suppose that a company wants its website to work seamlessly on mobile, tablet and desktop devices. Explain the concept of responsive web design and its importance. Describe the key CSS techniques used to achieve responsiveness, and explain how CSS Flexbox (or) Grid can be used to design adaptive layouts. Illustrate your answer with a simple example layout and suggest improvements to enhance user experience across different devices.

A) Responsive web Design:

Responsive web Design is a web design approach that enables a website to automatically adjust its layout, images, and content according to the screen size and device being used, such as mobile phones, tablets, and desktops. This ensures that users have a consistent and user-friendly experience without needing separate websites for different devices.

Importance of Responsive web design:

- provides a better user experience on all devices.
- Reduces the need to create separate mobile and desktop websites.

- Improves website accessibility and readability.
- Enhances search engine optimization (SEO) as search engines prefer mobile-friendly websites.

Key CSS Techniques for Responsiveness:

1) Media Queries: Apply different CSS styles based on screen width (or device characteristics).

CSS:

```
@media (max-width: 768px) {  
  {  
    .container {  
      flex-direction: column;  
    }  
  }  
}
```

2) Flexible Layouts:

→ Use percentage (%) (or) viewport units (vw, vh) instead of fixed pixel values.

3) Flexible images:

→ Make images resize automatically.

4) Responsive Typography:

→ Use relative units like em, rem, or vw for font sizes.

Using CSS Flexbox for Adaptive layout:

Example layout:

HTML:

```
<div class="container">
```

```
<div class="box"> Home</div>
```

```
<div class="box"> About</div>
```

```
<div class="box"> Contact</div>
```

```
</div>
```

CSS:

```
.container {
```

```
display: flex;
```

```
gap: 10px;
```

```
}
```

```
.box {
```

```
flex: 1;
```

```
padding: 20px;
```

```
background: lightblue;
```

```
text-align: center;
```

```
}
```

```
@media (max-width: 768px) {
```

```
.container {
```

```
flex-direction: column;
```

```
}
```

```
}
```

Improvements:

→ Use touch-friendly buttons with adequate spacing.

→ Test the website on multiple devices and screen sizes to ensure compatibility.

2) Client-side validation:

A)

client-side validation is the process of checking user input in the browser before the form is submitted to the server.

Purpose of client-side validation:

- To ensure that the entered data is valid.
- provides immediate feedback to the user.
- Improves overall user experience.
- Minimizes server processing time.

Javascript Event handling in validation:

- Javascript uses events such as `onsubmit`, `oninput`, `onchange` and `onblur` to validate form fields.
- `onsubmit`: validates all fields before the form is submitted.
- `oninput`: checks input while the user is typing.
- `onblur` - validates a field when the user leaves it.
- `onchange`: validates when the value of a field changes.

Regular expressions role:

These are used to verify whether user input follows a specific pattern, such as email address (or) phone-number.

Example:

HTML:

```
<form onsubmit = "return validateForm()">
```

```
<input type = "text" id = "email" placeholder = "Email">
```

```
<input type = "Submit" value = "Submit">
```

```
</form>
```

```
</script>
```

```
function validateForm() {
```

```
    let email = document.getElementById("email").value;
```

```
    let pattern = /^[^s@]+@[^s@]+\.[^s@]+$/;
```

```
    if (!Pattern.test(email))
```

```
    { alert ("Invalid Email");
```

```
      return false;
```

```
    }
```

```
    return true;
```

```
}
```

```
</script>
```

Improvements to enhance usability and security:

→ Display clear error messages near input fields.

→ validate input as the user types.

→ Use HTML5 validation attributes (required, type = "email")

→ perform Server-Side validation in addition to client-side validation for better security.

3) A) CSS Box Model:

The CSS Box model describes how elements are displayed on a webpage. It consists of four components.

- Content: The actual text or image
- Padding: Space between the content and the border.
- Border: Surrounds the padding and content.
- Margin: Space outside the border

CSS Positioning techniques:

- Static - Default positioning
- Relative - positioned relative to its normal position.
- Absolute - positioned relative to the nearest positioned ancestor.
- Fixed - Stays fixed relative to the browser window.
- Sticky - Switches between relative and fixed based on scrolling.

Possible causes of layout breaking:

- Using fixed widths instead of flexible widths.
- Large images or content exceeding the screen size.
- Incorrect use of CSS positioning
- Lack of responsive design for different screen sizes.

Using Media Queries:

Media queries apply different CSS styles based on the

Screen size, allowing the layout to adapt to different devices.

Example:

CSS:

```
@media (max-width: 768px) {
```

```
  .Container {
```

```
    flex-direction: column;
```

```
  }
```

```
}
```

Best practices for a stable and responsive layout:

- Use flexible units such as %, em, rem, or vw instead of fixed pixels.
- Apply media queries for different screen sizes.
- Use CSS flexbox or grid for responsive layouts.
- Make images responsive using max-width: 100%.
- Test the layout on mobile, tablet, and desktop devices.

Conclusion:

The CSS Box model and positioning techniques are essential for designing structured web pages. Using responsive design techniques such as media queries, flexible layouts, and responsive images ensures that websites display correctly on mobile, tablet, and desktop devices.

4) Document object model:

The Document object Model (DOM) is a programming interface for HTML and XML documents. It represents as a webpage as a tree-like structure of objects, where each HTML element is treated as node.

Role of the DOM in web development:

- Allows Javascript to access and modify HTML elements.
- Enables dynamic updates to webpage content.
- Responds to user actions such as clicks and keyboard input.
- Improves interactivity and creates responsive web applications.

Javascript DOM manipulation:

Javascript uses methods such as `getElementById()`, `querySelector()`, `innerHTML`, `textContent`, and `appendChild()` to update webpage dynamically.

Example:

HTML:

```
<P id="message"> welcome! </P>
```

```
<button onclick="changeText()" click Here </button>
```

```
<script>
```

```
function changeText() {
```

```
    document.getElementById("message").innerHTML =  
        "welcome to web programming!";
```

```
}
```

```
</script>
```

Explanation:

When the user clicks the button, the Javascript function `changeText()` accesses the paragraph using `getElementById()` and changes its content using `innerHTML`, without refreshing the page.

Improvements to enhance performance and experience:

- update only the required DOM elements instead of reloading the entire page.
- minimize the unnecessary DOM manipulations to improve performance.
- optimize Javascript code and cache frequently accessed DOM elements.
- Display loading indicators when fetching data from the server.